



Optimizing Biomedical Equipment Management Through a Business Intelligence Platform

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Abstract. This report presents the development and application of a Business Intelligence (BI) dashboard to improve healthcare management and decision making in medical facilities. Clinical Engineering and the efficient management of biomedical equipment are fundamental to ensure the quality of medical care and patient safety. Patient safety is of utmost relevance, and corrective and preventive maintenance of biomedical equipment is crucial to avoid adverse events or diagnostic errors.

The dashboard was meticulously designed to clearly and concisely present relevant information obtained from a thorough survey of medical equipment in key hospitals. The flexibility of the tool allows it to be adapted to various needs and geographic regions.

The accuracy and efficiency of the dashboard depend directly on the veracity and quality of the data collected, as well as on the regular updating of the database to reflect changes and evolutions in the healthcare system. The results obtained through the dashboard provide a comprehensive view of the distribution of hospitals, available medical equipment and bed characteristics in each region. This information is of great value to identify areas for improvement and optimization of resources in hospital management.

The health facility-oriented dashboard is positioned as a strategic tool to improve management, informed decision making and operational efficiency in healthcare. With careful implementation and accurate data, this tool contributes to improving the quality of healthcare and the well-being of the community at large.

Keywords: Clinical engineering · dashboard · business intelligence (BI) · biomedical equipment management

1 Introduction

Clinical Engineering plays an essential role in the improvement of medical care and patient care. In this context, it is necessary to perform the efficient management of biomedical equipment found in health institutions, with the aim of ensuring the quality of care and patient safety. This equipment has a fundamental role in diagnosis, treatment,

and rehabilitation, so its proper functioning is essential. Patient safety is one of the most important issues, so it is important to carry out corrective and preventive maintenance to ensure the effectiveness and safety of these devices, otherwise they could cause adverse events or diagnostic errors.

“In the field of public health there is a hierarchy of concepts: the data is the most primary unit that, by working with it, generates an indicator; this, once analyzed, generates information that, after being interpreted, generates knowledge.... This spiral of evidence production that ends in the action taken in the field of public health is the basis of what has been called “evidence-based public health” [1] It is of utmost importance to comply with the norms and regulations related to biomedical equipment to ensure that health facilities meet the required quality and safety standards. The efficient management of medical technology contributes to compliance with the quality standards established by the National Ministry of Health, through the National Program for Quality Assurance in Medical Care (PNGCAM) [1], as well as to the optimization of resources.

In the context of biomedical technology management, it is crucial to consider the need to replace equipment when it reaches the end of its lifespan. This technological replacement process must be carefully planned and scheduled in advance, in order to avoid unexpected expenses and ensure efficient management of available resources.

Adequate maintenance and efficient management ensure that equipment is available and working properly when needed, avoiding interruptions in treatment.

In Argentina, provincial governments have full autonomy in the implementation of public health policies and are primarily responsible for the provision of health care services in their jurisdictions. This decentralization implies that each province has the power to make decisions and establish strategies based on the needs and realities of its population.

Each provincial government manages its own resources and budget, which gives them the flexibility to prioritize health actions in accordance with local demands and needs and in compliance with the obligations established for public hospitals under decentralized management [2]. Regarding biomedical technology, this autonomy implies that provincial governments are responsible for the acquisition, maintenance, and renewal of biomedical equipment in the health facilities under their jurisdiction.

“Big data is poised to revolutionize health care, and performance dashboards can be an important tool to manage big data innovations. Dashboards show the progress being made and provide critical management information about effectiveness and efficiency. However, performance dashboards are more than just a clear and straightforward representation of performance in the health care context. Instead, the development and maintenance of informative dashboards can be more productively viewed as an interactive and iterative process involving all stakeholders” [3]. In this context, the need to optimize the management of biomedical equipment arises and, in response to this, the use of an interactive platform that is adaptable to the requirements of those responsible for technology management is proposed. This work presents the development of a Business Intelligence (BI) dashboard that allows a comprehensive and real-time visualization of relevant data, which facilitates decision making and the identification of areas for improvement in the management of equipment. This type of BI dashboard provides

a visual representation of the key indicators and data related to biomedical equipment in the form of graphs, tables, and interactive visual elements.

This study presents an innovative approach to optimize the management of biomedical equipment through a BI platform. This approach uses the analytical tool “Power BI” to develop a customized dashboard that optimizes the management of biomedical equipment, which facilitates decision making.

2 Materials and Methods

A thorough survey of medical equipment in key sectors of various hospitals was carried out in order to ensure the adequacy of the dashboard to the reality of medical technology in our country. This survey made it possible to adapt the platform to the different types of equipment and their specific characteristics, as well as to the particularities of the hospital infrastructure. In this way, a precise and customized configuration of the dashboard was achieved, ensuring its usefulness and effectiveness in the context of biomedical technology management in our country.

During the research process, several stages were carried out to collect and analyze information related to biomedical equipment and health facilities.

A. Characterization of health facilities and survey of biomedical equipment.

Data were collected from the Ministry of Health that provide detailed information on the general characteristics of the institutions [2] and the number of beds available in each of them [3]. In addition, the different bed distribution areas (such as inpatient, on-call and therapy) were identified in order to obtain a comprehensive view of hospital capacity.

Subsequently, visits were made to several hospitals in the province to carry out a thorough survey of medium and high complexity medical equipment. The equipment analyzed included monitors, mechanical ventilators, defibrillators, anesthesia machines, diagnostic imaging equipment and electrosurgical equipment, among others. Relevant data were recorded for each piece of equipment, referring to its characteristics and state of use.

B. Development of the dashboard.

Using the analytical tool “Power BI”, a dashboard was developed, designed to present the information in a clear and understandable manner. This was accomplished through a variety of data visualization techniques to effectively represent and communicate in real time the relevant patterns, trends and findings derived from the data collected.

Through the combination of graphs, charts and other interactive visual elements, it was possible to facilitate the information understanding and analysis. The dashboard in question is integrated with the following modules (Fig. 1):

In the first module, a thorough analysis is carried out on the distribution of hospitals according to the different health regions, in addition to examining the predominant equipment and brands in these institutions. The services with the largest amount of medical equipment available are also identified.

The second module presents a detailed study on the age and state of operation of medical equipment. Additional information is included highlighting the main companies

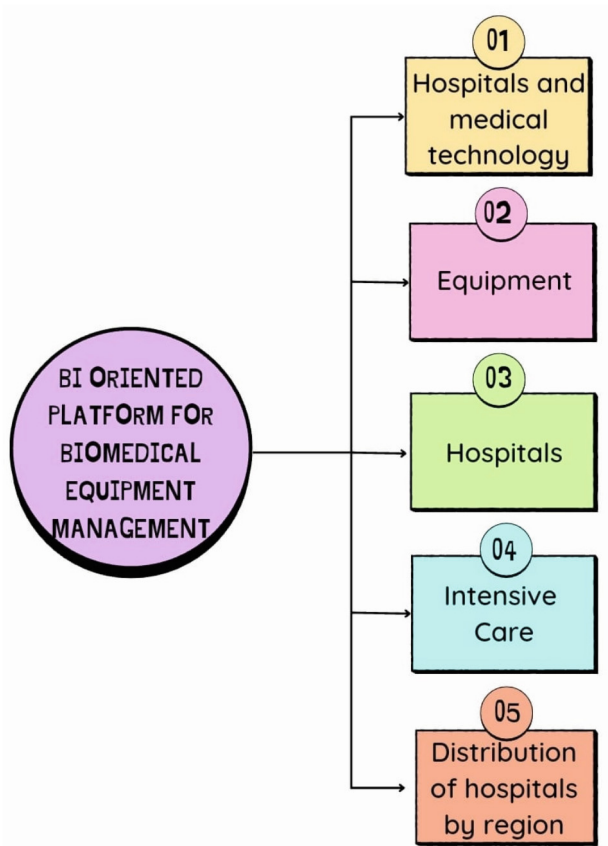


Fig. 1. Dashboard modules

responsible for the maintenance of this equipment, which is relevant to ensure its optimal performance.

The third module focuses on a comprehensive analysis of hospital characteristics, covering crucial data such as the number of available beds and their distribution according to health regions and hospital services. A distinction is also made between the types of beds available, making a difference between acute and chronic beds, and hospitals with and without inpatient services are identified.

The fourth module presents specific information on the availability of equipment and beds in intensive care units. From these data, essential indicators are developed to establish the ratio of mechanical ventilators and monitors per bed, which is crucial for improving care and quality of medical care.

Finally, the fifth module highlights the presentation of a map that visualizes the geographical location of hospitals in the province of Buenos Aires, using distinctive colours to represent the different health regions. It is important to note that, although the

province of Buenos Aires has been taken as an example, this methodology is applicable to any national territory.

This methodological approach provides a complete view of the distribution of hospitals, available equipment and bed characteristics in each region. The information obtained is of great value for management and decision making in the health sector, making it possible to improve the efficiency and effectiveness of the medical services.

3 Outcomes

Following the data analysis and the implementation of the advanced visualization tool, a dashboard reflecting the results obtained from the mentioned modules was obtained.

This dashboard has been meticulously designed to clearly and concisely present the relevant information derived from each of the analysis modules.

In Fig. 2 and Fig. 3, a visualization of the most significant patterns, trends and findings related to different key aspects of the health system is intuitively presented. These figures provide a graphic representation of the distribution of hospitals, the available medical equipment with its general characteristics, the services with the most biomedical technology, the predominant brands in the healthcare field and the main companies in charge of preventive and corrective maintenance of the equipment.

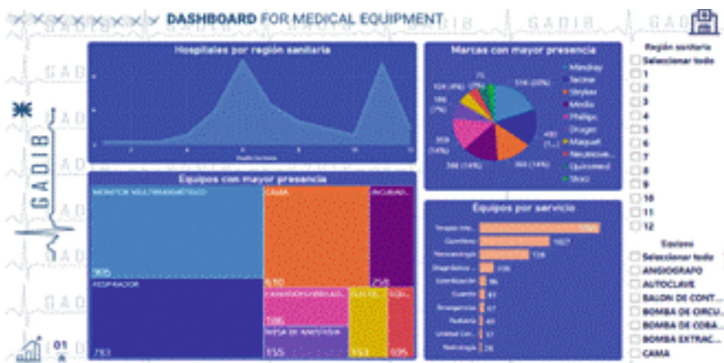


Fig. 2. Module N°1: “Hospitals and medical technology”

Figure 4 shows relevant information on the number of acute and chronic beds, as well as the medical equipment associated with intensive care units. Among the data obtained, it can be seen the percentage of acute and chronic beds, the distribution of total beds by region and the services with a predominance of acute beds.

Figure 5, on the other hand, maintains the focus on the intensive care service, detailing the equipment operating in this area and the number of beds available, which are differentiated according to the type of critical care service. Also, indicators have been designed to reflect the numbers of monitors and mechanical ventilators per bed in this type of sector, which is essential to ensure optimal care of critically ill patients.

Finally, Fig. 6 shows the distribution of hospitals by health region in the province of Buenos Aires, allowing a clear visualization of the distance between the different

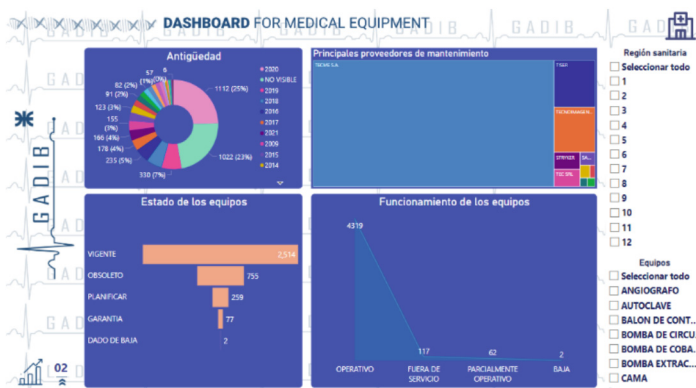


Fig. 3. Module N°2: "Equipment"

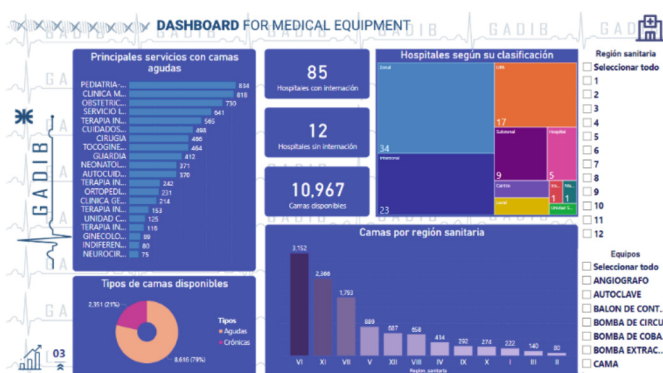


Fig. 4. Modules N°3: "Hospitals"

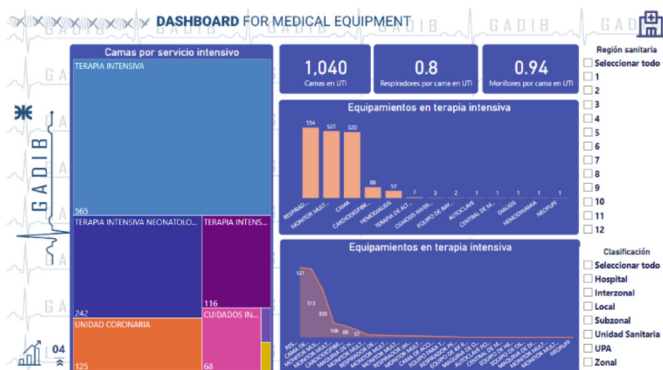


Fig. 5. Module N°4: "Intensive care"

hospitals. This geographical representation is valuable for identifying areas with greater or lesser medical coverage and improving the strategic planning of health services in the region.

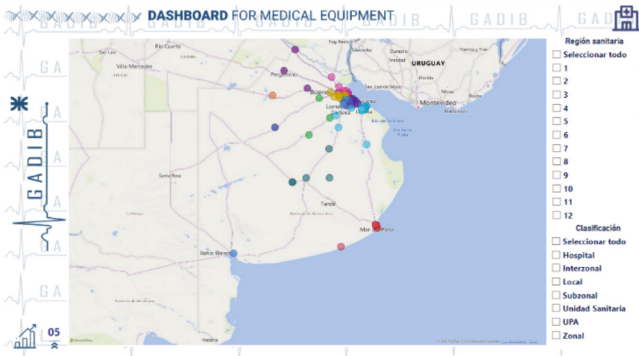


Fig. 6. Modules N°5: "Distribution of hospitals by region".

As we can see, the dashboard is a valuable data presentation resource that provides a comprehensive view of the current situation of the health system, facilitating the identification of areas for improvement, strengths, and opportunities for the optimization of hospital services and the efficient management of resources in each health region.

4 Discussion

“A dashboard is one of technology’s business intelligence tools that provides awareness, trending and comparisons of both operations and finances. A well-defined and comprehensive dashboard can translate data into information that supports making knowledge-based decisions.” [6]. The dashboard that has been presented has the ability to be adapted to various needs and regions. The most crucial stage in the creation of the dashboard is data collection, since accurate and verifiable information is required to ensure reliable results. Thus, the effectiveness of the dashboard will depend directly on the veracity and quality of the data with which it is constructed.

It is also important to keep the dashboard database updated as changes and developments occur in the health system. The regular information update ensures that the dashboard remains a valuable and useful tool for informed decision making and efficient resource management in the health field.

In conclusion, the flexibility and adaptability of the dashboard allows its application in different contexts, as long as reliable data are available. The accuracy of the results will be directly linked to the quality of the information collected and its constant updating.

5 Conclusions

The health facility-oriented dashboard is a valuable tool for improving management and decision making in this critical area. Its flexibility and adaptability to different needs and regions make it suitable in different geographical contexts, which makes it an important resource for health authorities and managers.

The collection of reliable data is the most relevant stage in the production of the dashboard, since the quality of the information will determine the accuracy and reliability of the results obtained. With accurate data, dashboard analysis can provide a clear picture of the distribution of hospitals, available medical equipment, bed characteristics and other aspects crucial for informed health care decision-making.

In conclusion, the health facility-oriented dashboard is positioned as a strategic tool for successful management in this field. It provides a comprehensive view of the situation and allows the identification of areas for improvement, priorities for investment in medical equipment and services, and optimization of resources. With careful implementation and a solid database, the dashboard can improve the quality of medical care, operational efficiency and ultimately contribute to a more effective health system oriented to the well-being of the community.

References

1. Health indicators. Conceptual and operational aspects. https://iris.paho.org/bitstream/handle/10665.2/49058/9789275320051_spa.pdf?sequence=5&isAllowed=y
2. Resolution 432/1992 secretary of health. <https://e-legis-ar.msal.gov.ar/hdocs/legisalud/migracion/pdf/17554.pdf>
3. National Decree 939/2000. <http://servicios.infoleg.gob.ar/infolegInternet/anexos/60000-64999/64697/texact.htm>
4. Big data health care innovations: performance dashboarding as a process of collective sensemaking: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8905474/>
5. Health facilities. <https://catalogo.datos.gba.gob.ar/dataset/establecimientos-salud>
6. Acute and chronic beds. <https://catalogo.datos.gba.gob.ar/dataset/camas>
7. Dashboards: a required business management tool. <https://doi.org/10.2345/0899-8205-44.3.228>